1 Monitoring and Observability

1.1 Components

The monitoring stack was developed to monitor the Docker Swarm infrastructure and provide information about the performance and availability of the different services. Prometheus was used to collect and store metrics from different services. Information is collected every 15 seconds and and stored so that it can be analysed over time. Grafana is then used to visualise this information through dashboards. To collect metrics, two exporters were used. Node Exporter provides information about the servers, such as CPU usage, memory usage and system load, while cAdvisor collects information about the resources consumed by individual Docker containers. Furthermore, Blackbox Exporter was used to check the availability of different application services.

1.2 Deployment in Docker Swarm

Node Exporter and cAdvisor were deployed in global mode, which means that Docker Swarm runs one instance of each service on every node. This allows metrics to be collected from all the nodes. Prometheus and Grafana run on `balkan1` which is the manager node. Blackbox Exporter runs with two replicas distributed across different nodes. Prometheus uses Docker Swarm service discovery to automatically find the exporters running on the different nodes. This is useful because containers can be restarted or moved between nodes without having to manually update their addresses. The monitoring services communicate through a Docker overlay network, while Grafana is connected to the frontend network so that users can access it through the Traefik reverse proxy. Passwords are managed using Docker Swarm secrets instead of storing them directly in the configuration files. Furthermore, Docker volumes were configured for Prometheus and Grafana to preserve their information when containers are restarted. Prometheus stores the collected metrics, while Grafana stores its internal configuration and data.

1.3 Grafana Dashboards

Grafana was configured to automatically load its datasources and dashboards using configuration files. This means that the dashboards do not need to be manually created every time the monitoring stack is deployed. The main dashboard displays different metrics about the infrastructure. For each server, it shows CPU usage, memory usage, system load and whether Prometheus can collect metrics from it. Furthermore, cAdvisor provides information about the CPU and memory used by each individual container. The dashboard allows the user to select which nodes and containers to display. Container information is also organized using readable names, making it easier to identify the different services and their replicas. The dashboard refreshes every 15 seconds to display recent information about the infrastructure.

1.4 Application Availability Monitoring

In addition to monitoring resource consumption, it is also important to know whether the status of applications is healthy. For this reason, Blackbox Exporter was configured to perform different checks on the services deployed.The Core application is monitored by checking whether its HTTP endpoint returns a successful response. PostgreSQL connectivity is verified through a health endpoint in the Core application, which executes a simple database query. Redis is monitored by sending a PING command and checking whether it responds with PONG. Finally, the Atenea Moodle platform is monitored through its public HTTPS endpoint. Grafana displays the results of these checks. This allows problems with the applications to be identified without having to inspect each container individually.

1.5 Grafana Plugin

Grafana was also connected to the application's PostgreSQL database using its built-in PostgreSQL datasource plugin. This allows data from the database to be queried directly from Grafana. A separate database user was created with read-only permissions on the required application table. This prevents Grafana from modifying the application's data.

1.6 Results

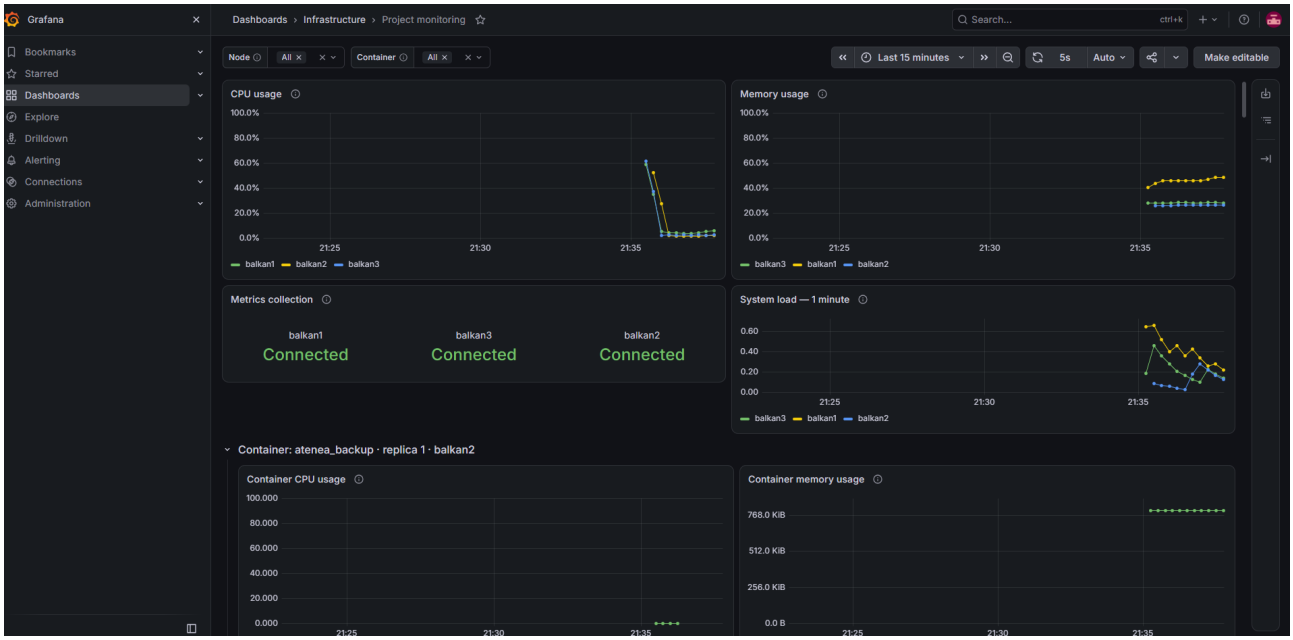


Figure 1: The dashbaord displays metrics for all nodes

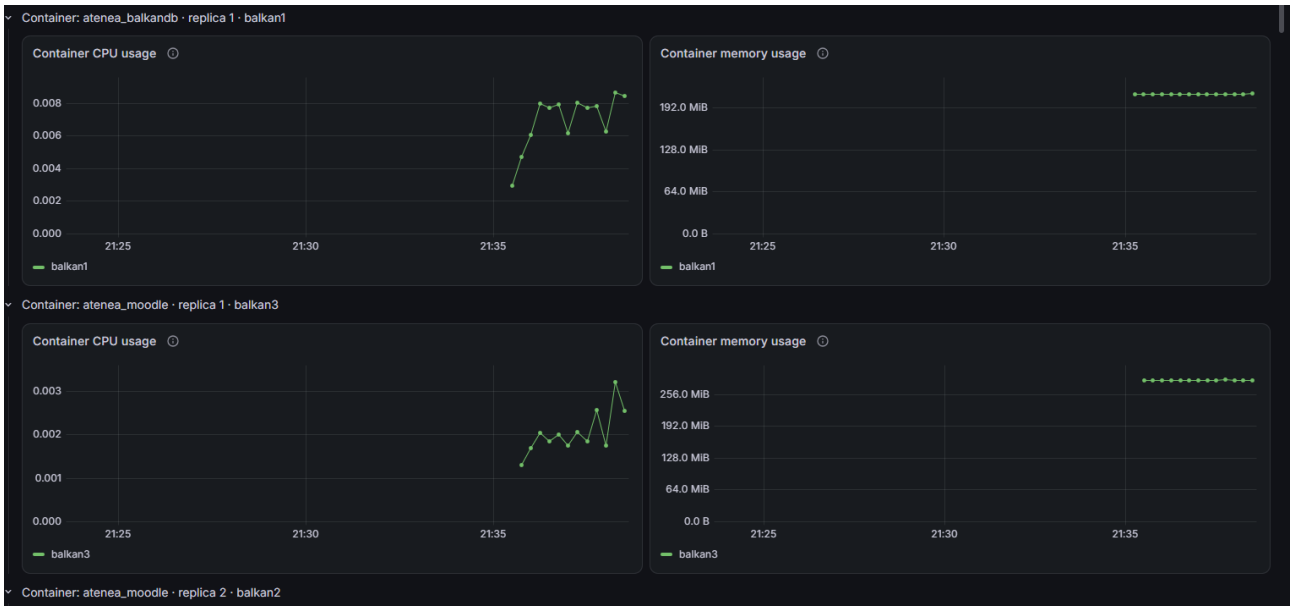


Figure 2: The dashboard displays metrics for individual containers



Figure 3: The dashboard displays the applications service checks

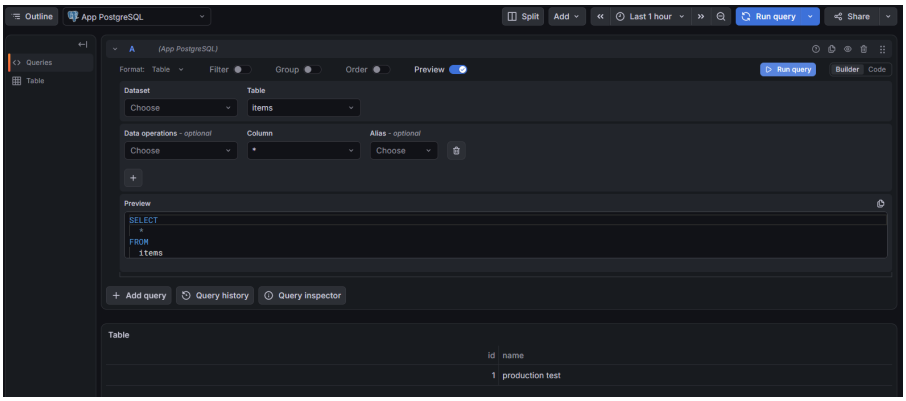


Figure 4: The dashboard allows to query the database